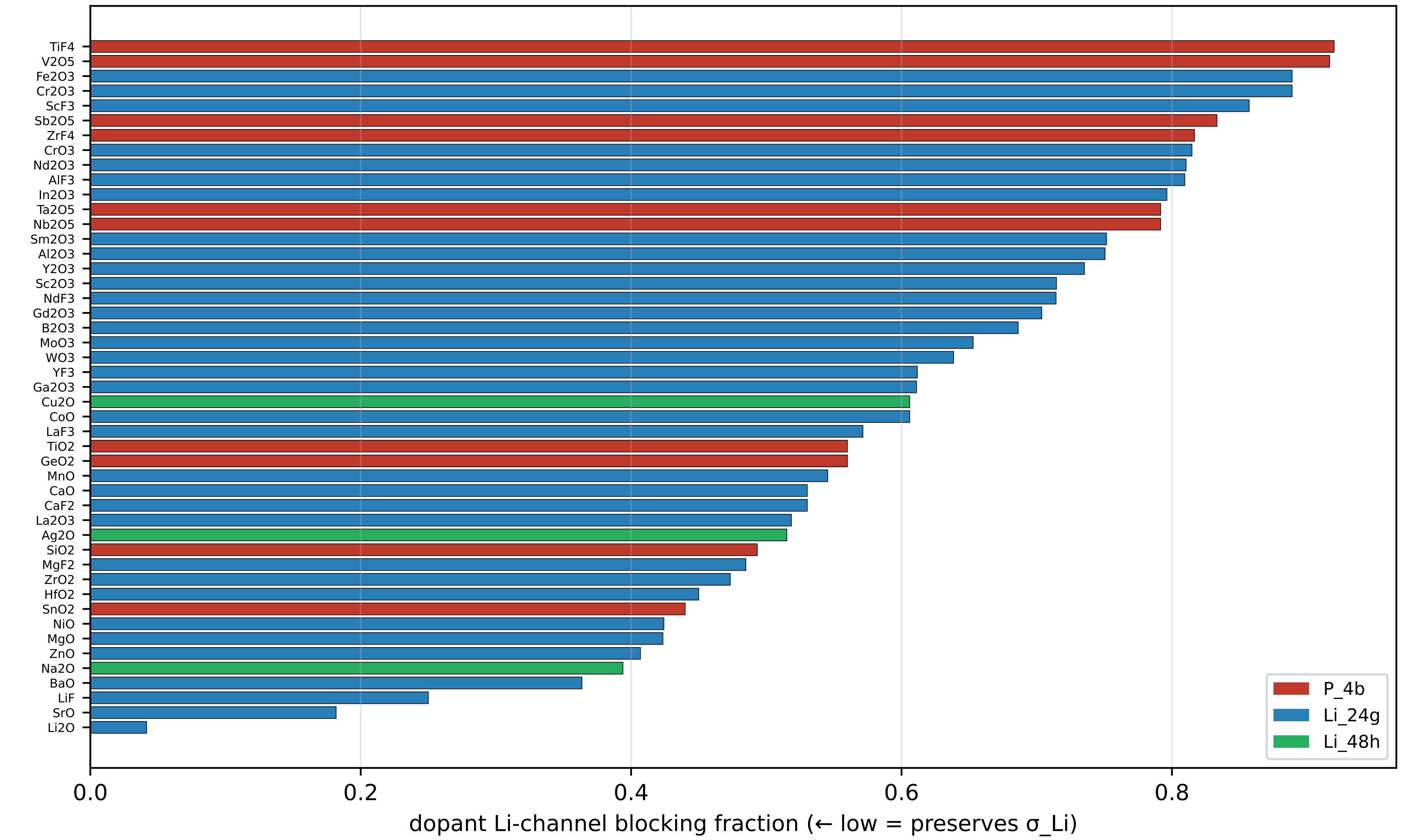
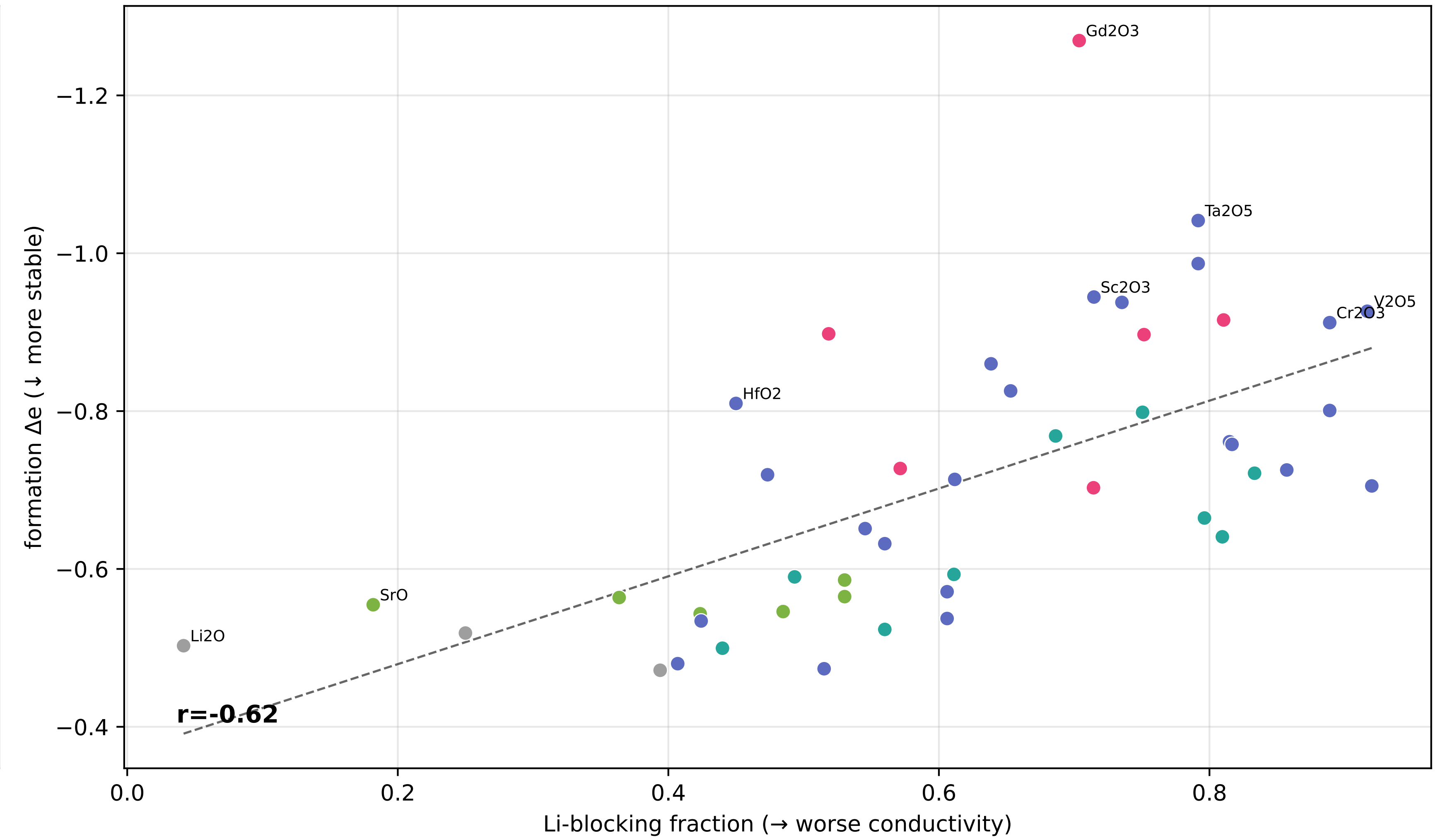
Cascade v23 — Li-mobility + site preference (the SE-critical dimension)

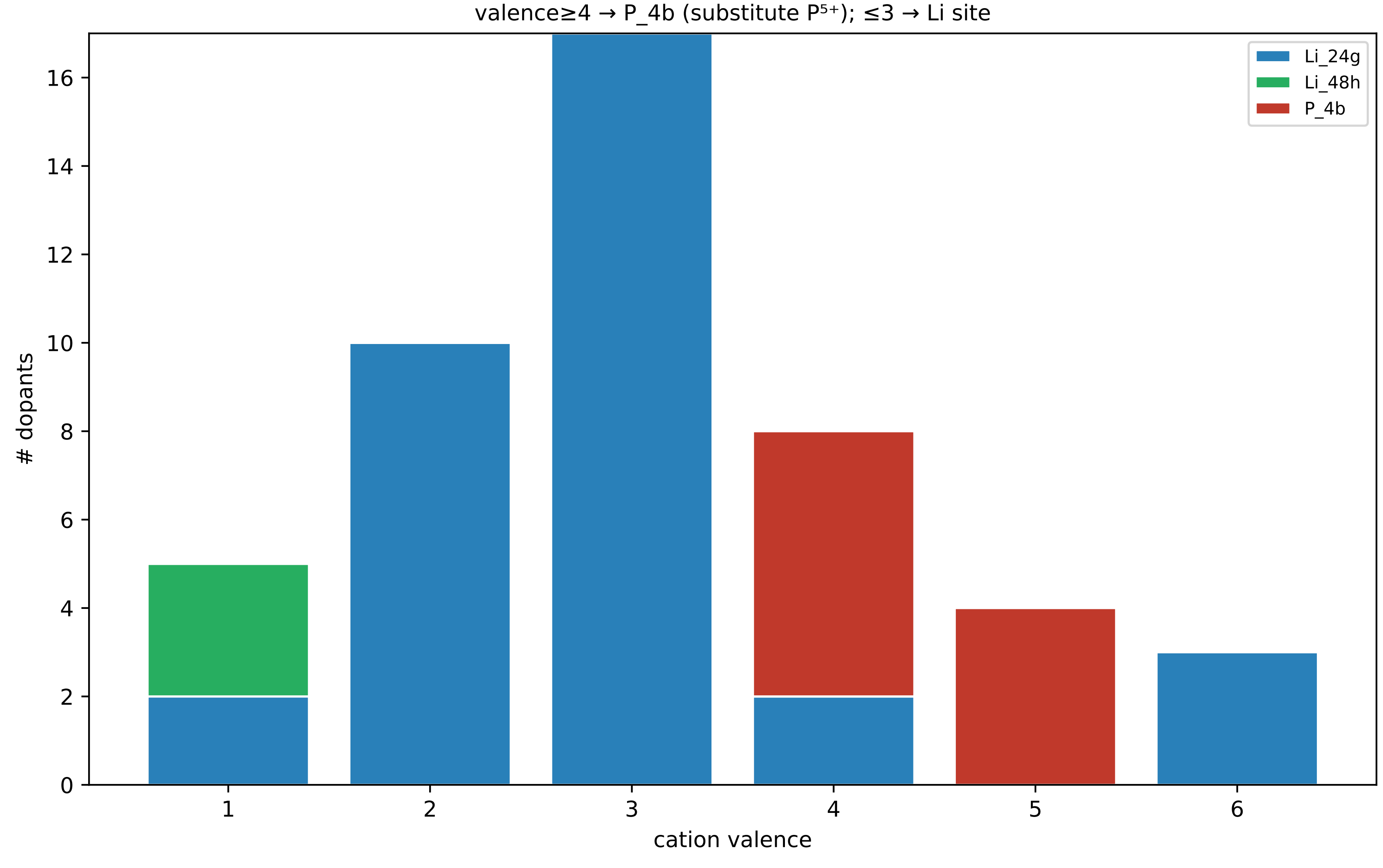
(A) Li-blocking ranking — low=good conductor. color=cation site



(B) ★ Stability ↔ Mobility TRADE-OFF
most-stabilizing dopants block Li most (lower-right tension); HfO2 escapes it



(C) Site preference = doping mechanism
valence≥4 → P_4b (substitute P⁵⁺); ≤3 → Li site



(D) TRUE 4-objective ranking (now incl. Li-mobility)

4-objective score = 0.30 stable + 0.25 oxidation + 0.25 Li-mobile(low-block) + 0.20 soft

rank	dopant	score	de	ox	blk	E	site
1	Gd2O3	0.65	-1.27	2.12	0.70	42	Li_24g
2	Cr2O3	0.61	-0.91	2.36	0.89	37	Li_24g
3	Sc2O3	0.61	-0.94	2.36	0.71	45	Li_24g
4	B2O3	0.56	-0.77	2.32	0.69	41	Li_24g
5	WO3	0.55	-0.86	2.14	0.64	39	Li_24g
6	Ga2O3	0.53	-0.59	2.36	0.61	41	Li_24g
7	In2O3	0.50	-0.66	2.36	0.80	42	Li_24g
8	Li2O	0.50	-0.50	2.14	0.04	49	Li_24g
9	MoO3	0.50	-0.83	2.14	0.65	43	Li_24g
10	Y2O3	0.49	-0.94	2.28	0.74	53	Li_24g
11	Al2O3	0.46	-0.80	2.14	0.75	43	Li_24g
12	Nb2O5	0.46	-0.99	2.06	0.79	46	P_4b
13	HfO2	0.45	-0.81	2.00	0.45	47	Li_24g
14	MgF2	0.44	-0.55	2.14	0.48	43	Li_24g